

SUPERCAPACITORS: WILL THEY REPLACE BATTERIES?

A supercapacitor along with conventional battery can be a better electrical energy storage device than the battery alone



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Delivery of electrical power is becoming an increasingly complex task for engineers and policy makers alike. The electricity systems are poised to change in order to accommodate increasing levels of distributed storage and utilisation. These complexities are being recognised internationally for alternating-current systems. But there is need to look beyond it as most consumer applications use direct-current supply and hence the smart energy delivery system concept is required to be extended to low- and medium-power direct-current systems.

Electrical energy storage (EES) refers to the process of converting electrical energy from a power source

into an electrical or non-electrical form that can be stored for converting back into electrical energy as per requirement. This process enables electricity to be produced at times of either low demand, low generation cost or from intermittent energy sources, and used at times of high demand, high generation cost or unavailability of other generation means.

Smart EES has numerous applications in portable devices, transport vehicles, stationary energy resources, etc. EES becomes more important in situations where we produce large amount of electricity from time-dependant energy sources like wind and solar. Smart EES has to consider generation, transmission/distribution, energy management system and renewable energy. Research should

focus on energy storage devices, particularly pulse power devices such as supercapacitors.

Types of smart EES

The various smart EES systems can be categorised based on their function and form. High-power, low-energy devices can be categorised by function, whereas high-energy, low-power systems can be categorised by form. Although electricity is not easy to store cheaply, it can be easily stored in other forms and converted back into electricity when needed.

Storage technologies for electricity can also be classified by the form of storage:

Electrical energy storage. Electrostatic energy storage includes capacitors and supercapacitors, and magnetic/current energy storage includes superconducting magnetic energy storage.

Mechanical energy storage. Kinetic energy storage includes fly-wheel-pumped hydroelectric storage, compressed-air energy storage and potential energy storage systems.

Chemical energy storage. Chemical energy is stored in batteries such as lead-acid batteries, nickel-cadmium (NiCd) batteries, sodium sulfur (NaS) batteries, sodium nickel chloride batteries, lithium-ion batteries and fuel cells.

Thermal energy storage. Low- or high-temperature energy is stored through devices like solar fuels.

Of these, chemical energy storage devices are the most suitable for DC power storage. Supercapacitors make